Problem Statement

ABC is India based entertainment production company with a focus on Northeast and East Indian cinema. They wanted a highly available and reliable storage solution for their on prem data. The client wanted the data to be reliable, highly available, secure, and persistent and also wanted to have data on the storage to be accessible from all the servers. The issues with the existing infrastructure are mentioned below:

1) They were using NAS storage which had limitations in scalability.

2) The client wanted a complete application to run in Aws cloud with storage in the cloud to keep his files in sync with servers so that he has centralized storage of data

3) There were security issues in the existing infrastructure and no encryption at rest or transit.

4) Manual intervention was needed to change the storage type and transfer files to infrequent storage.

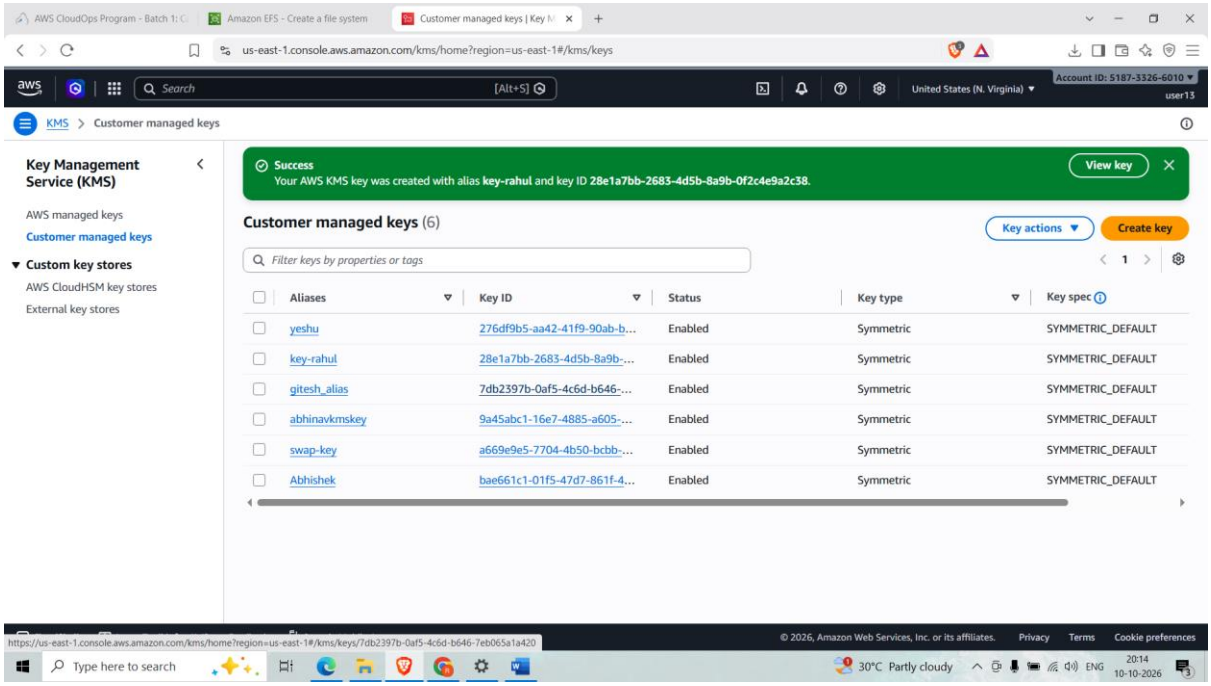
5) The client is unable to scale up the infrastructure due to high capital costs for new hardware.

6) Need for low-cost storage options for both frequent and infrequent data.

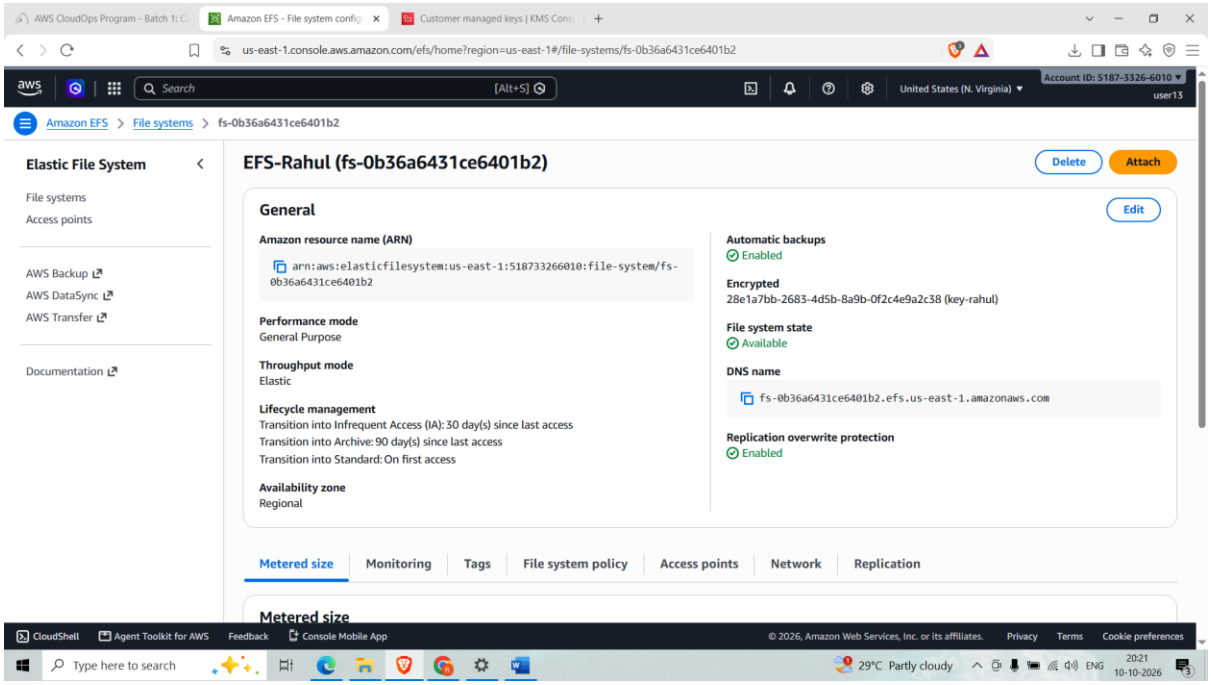
7) Highly available, secure, and persistent shared File system in AWS cloud with EFS.

1- Creating an EFS

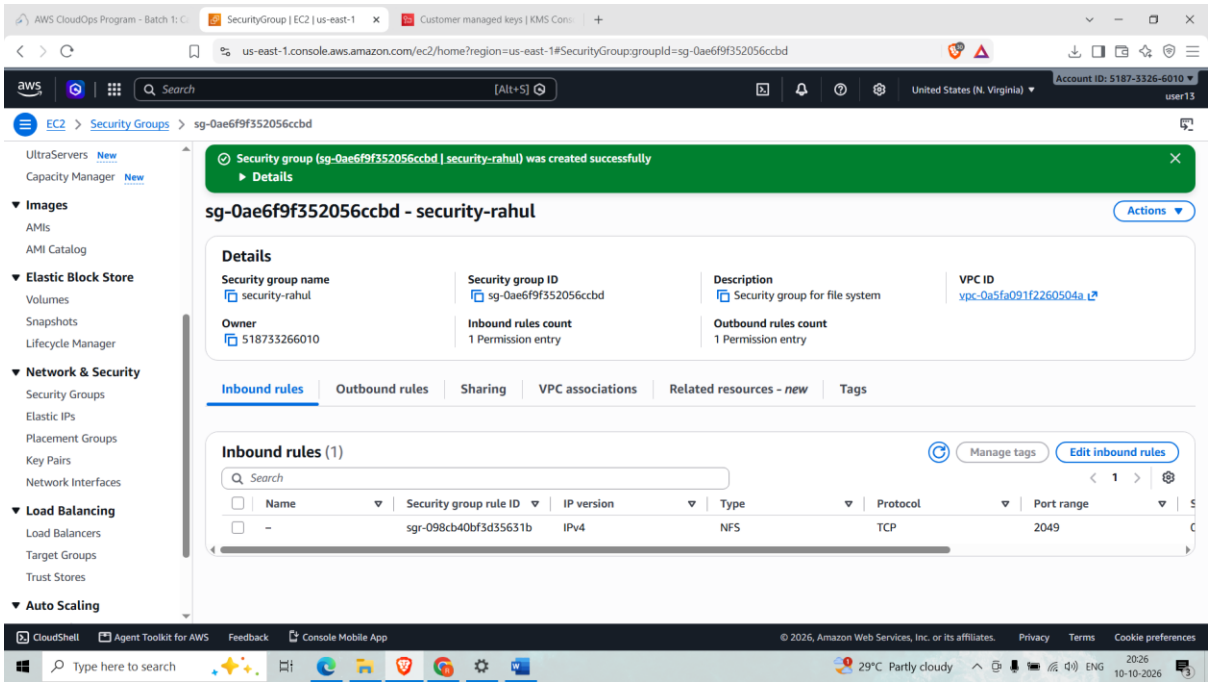
1.1 creating kms



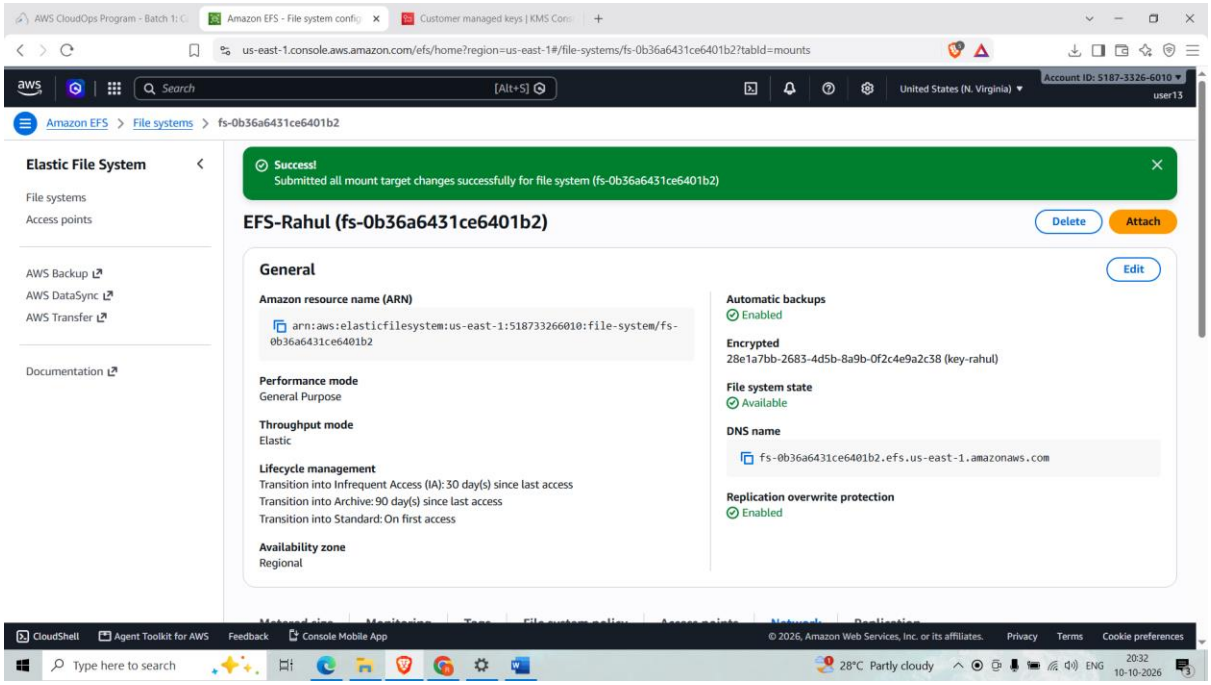
1.2 EFS file created



1.3 Creating a security group (Security group for file system)

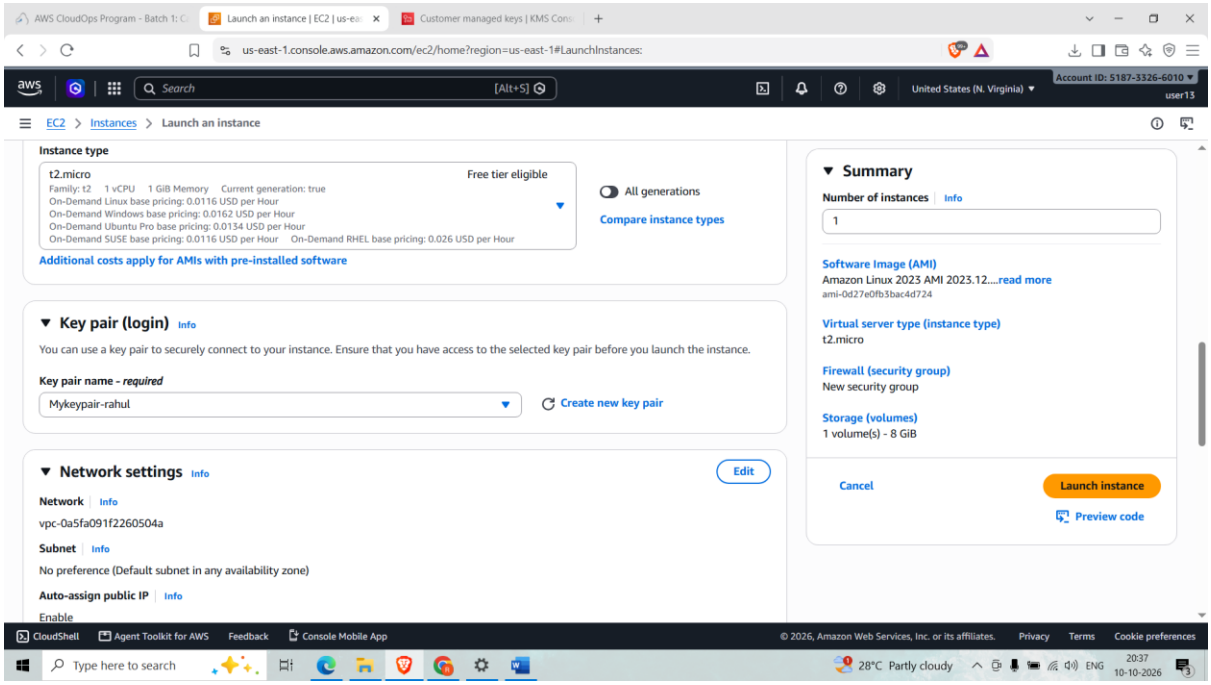


1.4 security group added to file system

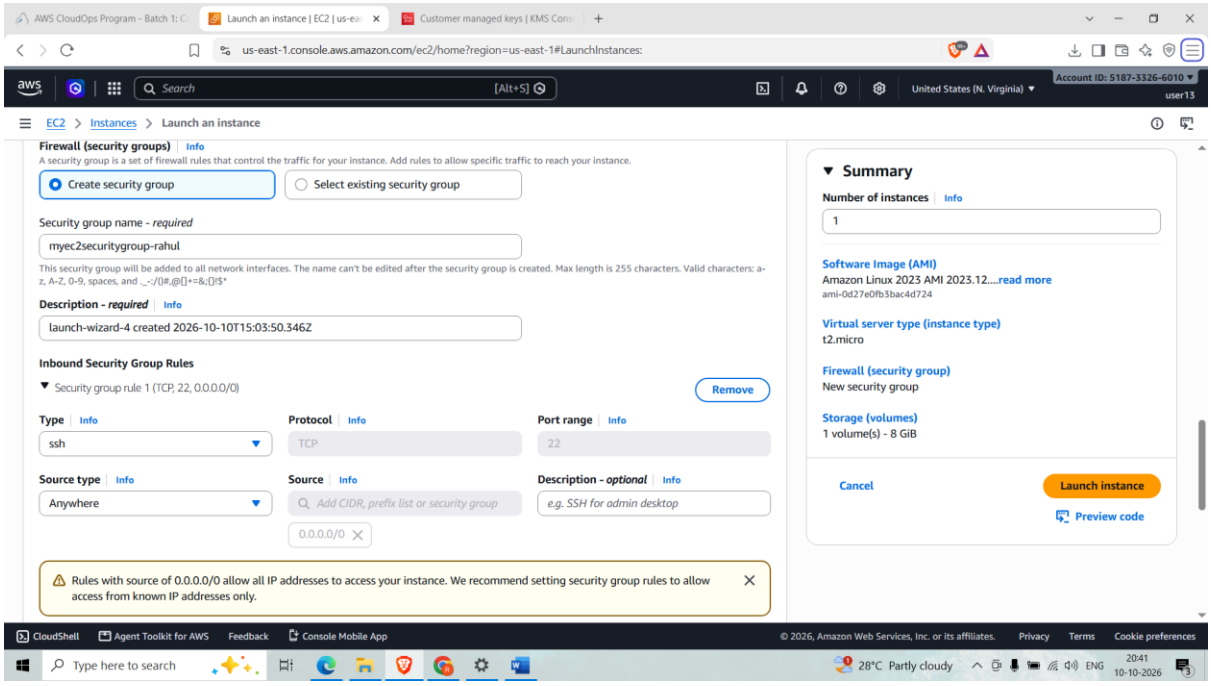


1.5 Creating first EC2 as per requirement

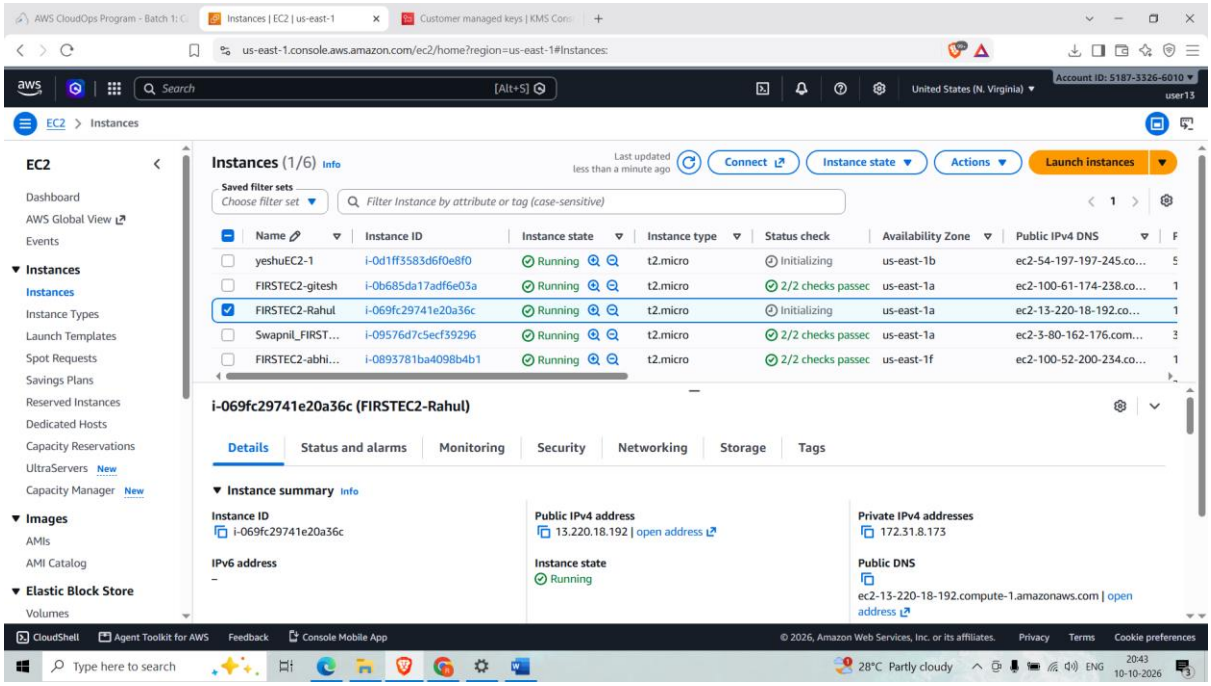
1.5.1 keypair created



1.5.2 created a firewall for ec2

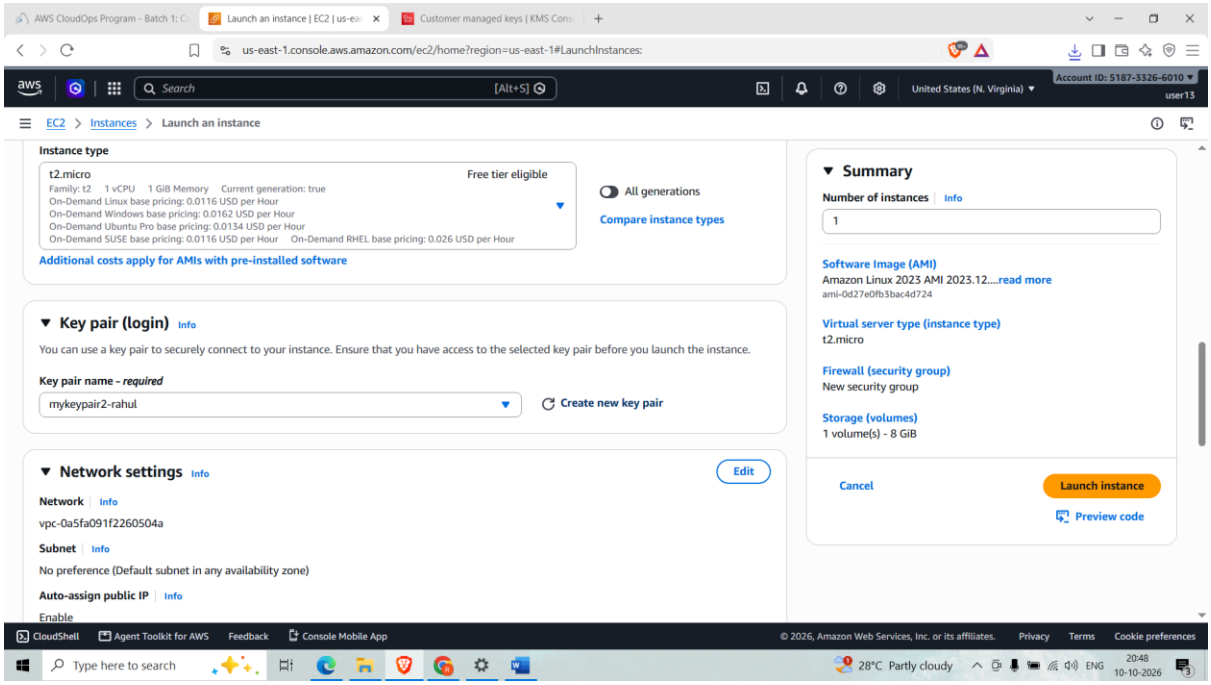


1.5 first EC2 launched successfully

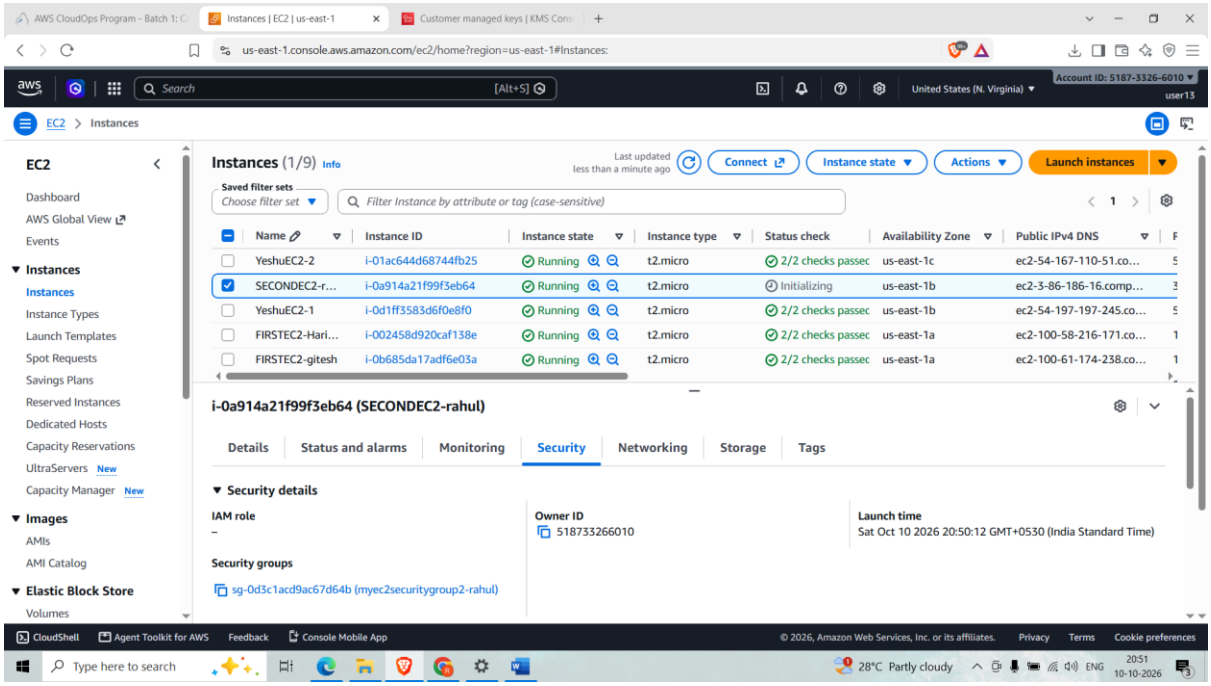


1.6 Creating second EC2 as per requirement

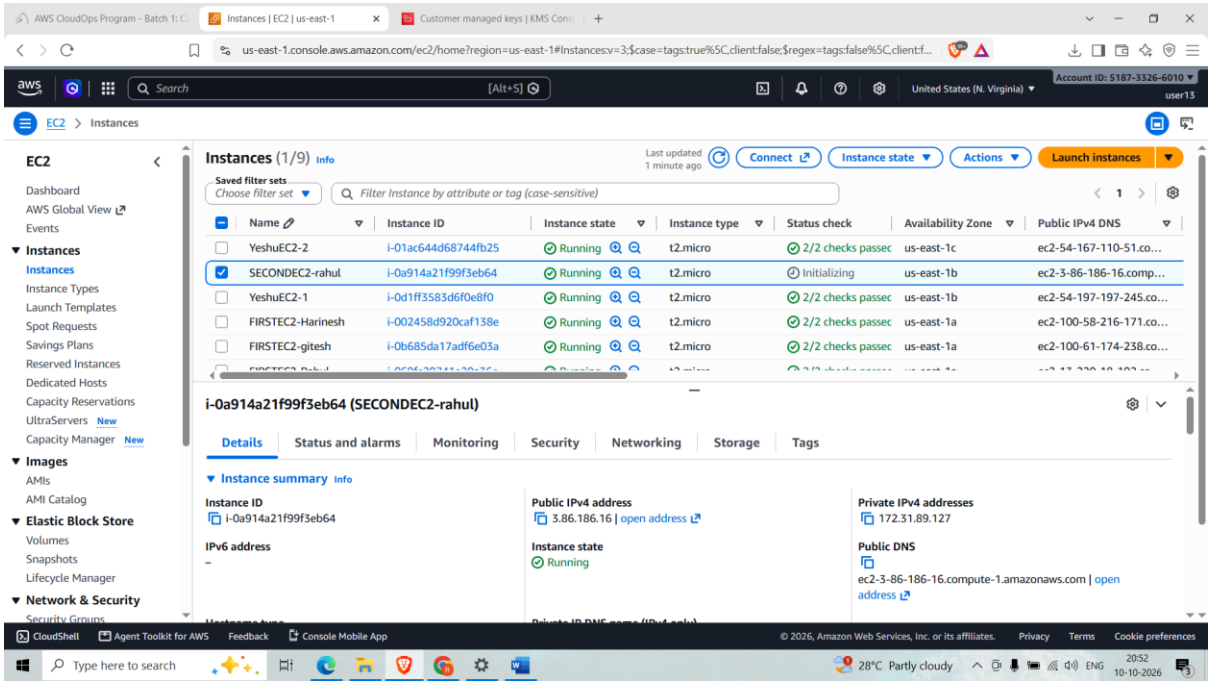
1.6.1 new keypair created for second ec2 server



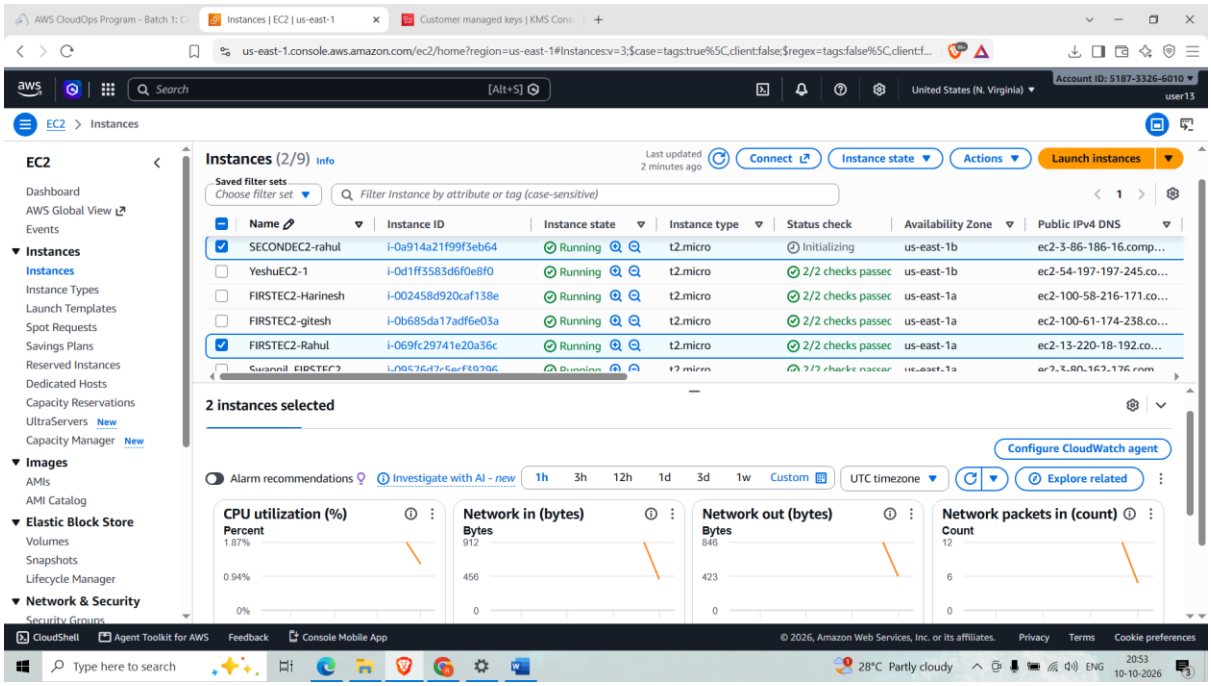
1.6.2 created a second firewall for second ec2



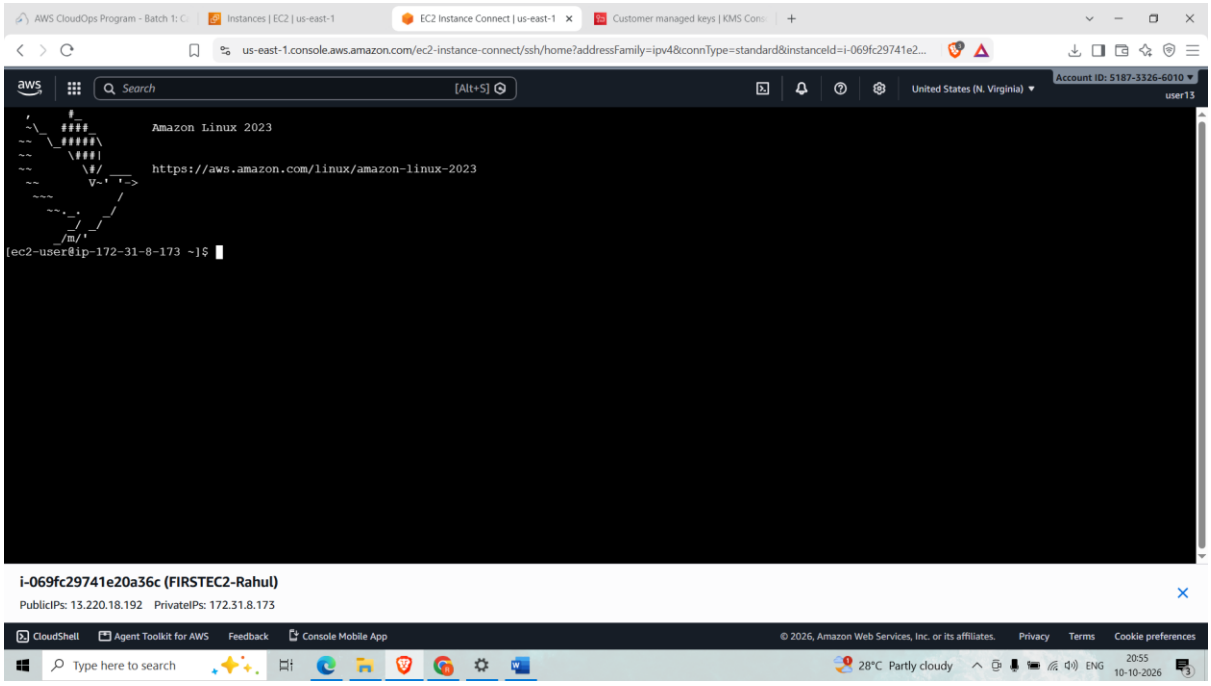
1.6 SUCCESSFULLY CREATED second EC2



2 instances created at different time zone



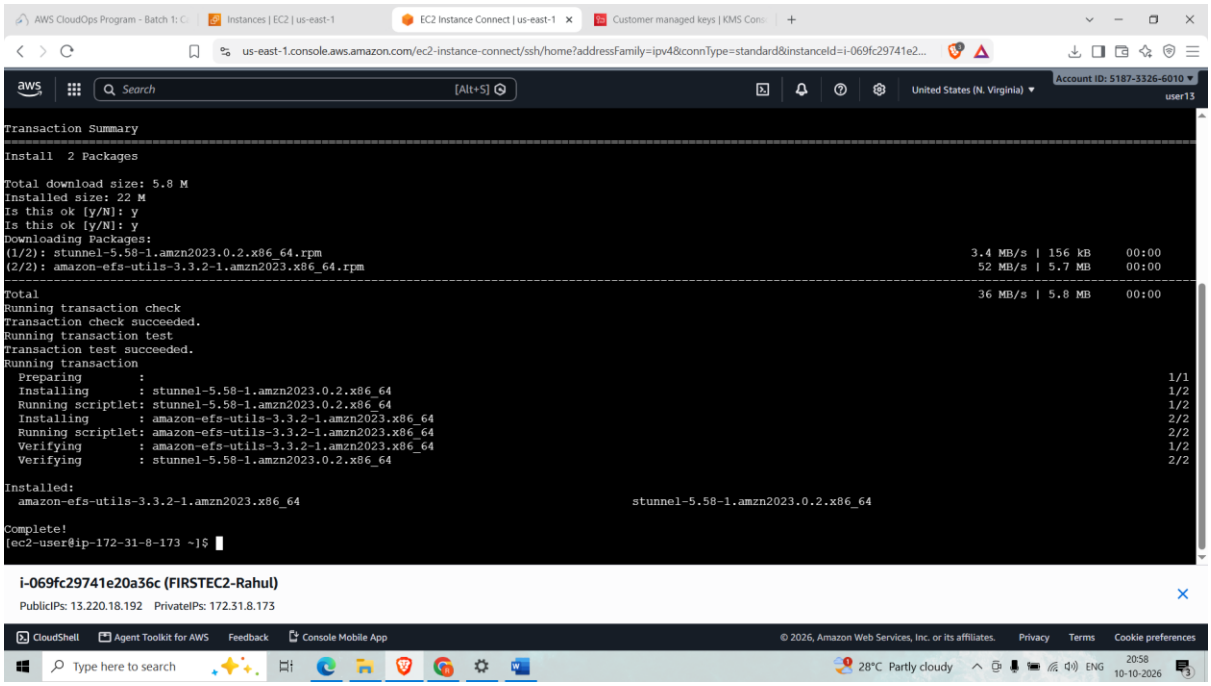
1.7 Connected the first ec2 instance (FIRSTEC2-Rahul)

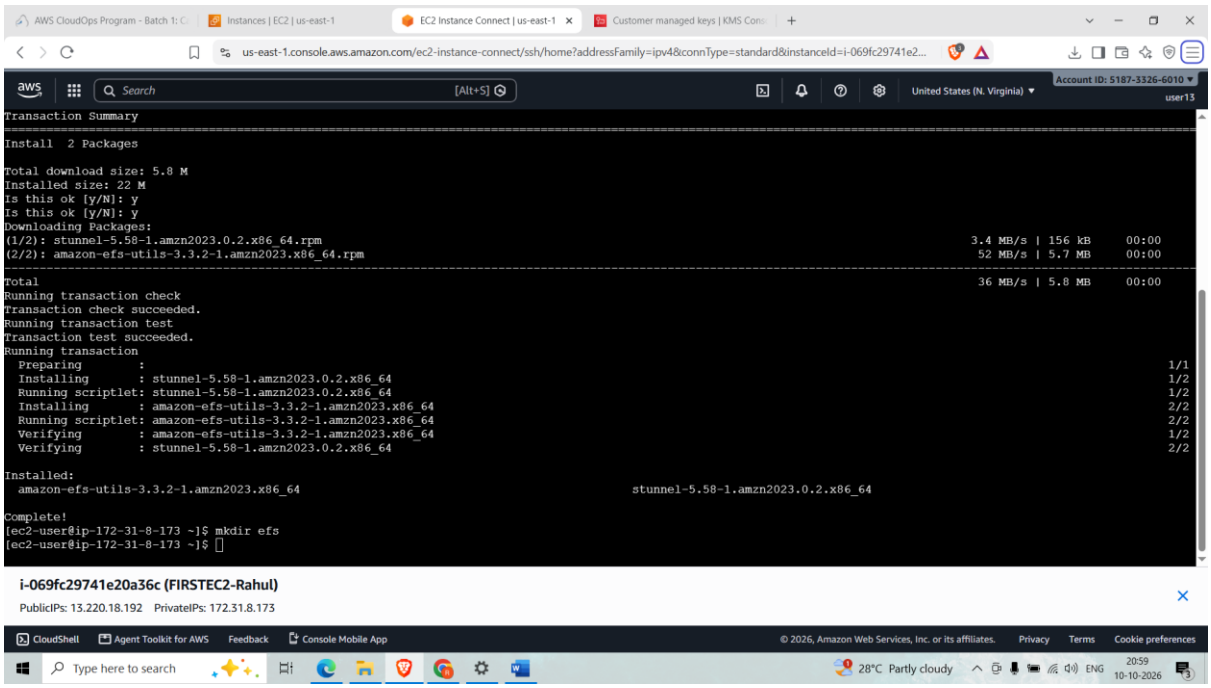


1.7.1 installing amazon efs utilities (creating efs server)

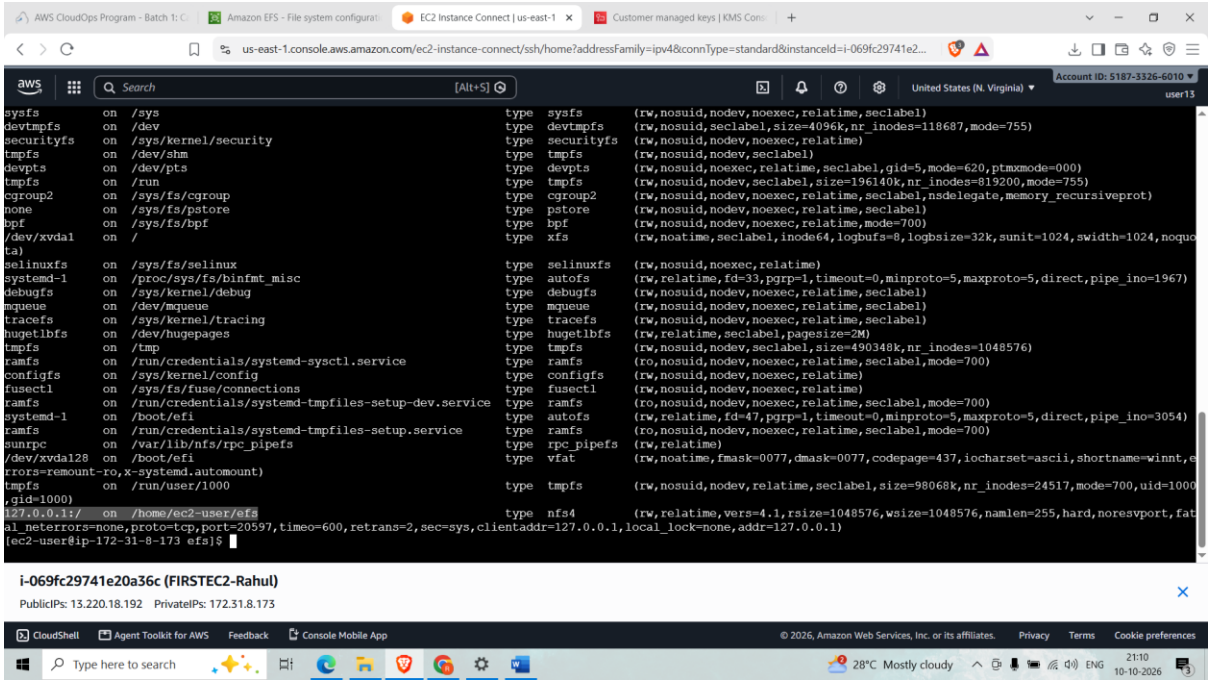
Command used : sudo yum install amazon-efs-utils

: mkdir efs


AWS CloudOps Program - Batch 1: C...
Instances | EC2 | us-east-1
EC2 Instance Connect | us-east-1
Customer managed keys | KMS Cons
us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipv4&connType=standard&instanceId=i-069fc29741e2...
Search
[Alt+S]
United States (N. Virginia)
Account ID: 5187-3326-6010
user13
Transaction Summary
Install 2 Packages
Total download size: 5.8 M
Installed size: 22 M
Is this ok [y/N]: y
Is this ok [y/N]: y
Downloading Packages:
(1/2): stunnel-5.58-1.amzn2023.0.2.x86_64.rpm 3.4 MB/s | 156 kB 00:00
(2/2): amazon-efs-utils-3.3.2-1.amzn2023.x86_64.rpm 52 MB/s | 5.7 MB 00:00
Total 36 MB/s | 5.8 MB 00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
Preparing : 1/1
Installing : stunnel-5.58-1.amzn2023.0.2.x86_64 1/2
Running scriptlet: stunnel-5.58-1.amzn2023.0.2.x86_64 1/2
Installing : amazon-efs-utils-3.3.2-1.amzn2023.x86_64 2/2
Running scriptlet: amazon-efs-utils-3.3.2-1.amzn2023.x86_64 2/2
Verifying : amazon-efs-utils-3.3.2-1.amzn2023.x86_64 1/2
Verifying : stunnel-5.58-1.amzn2023.0.2.x86_64 2/2
Installed:
amazon-efs-utils-3.3.2-1.amzn2023.x86_64 stunnel-5.58-1.amzn2023.0.2.x86_64
Complete!
[ec2-user@ip-172-31-8-173 ~]$
i-069fc29741e20a36c (FIRSTEC2-Rahul)
PublicIPs: 13.220.18.192 PrivateIPs: 172.31.8.173
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
Type here to search
28°C Partly cloudy
ENG
20:58
10-10-2026

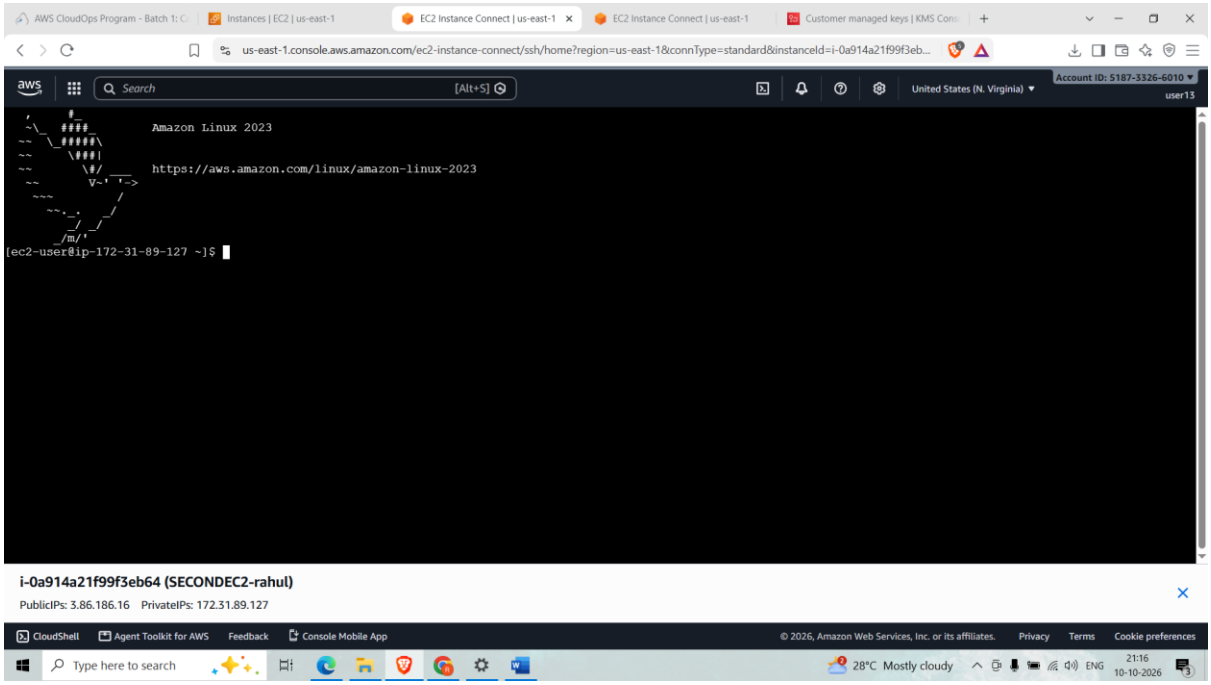

AWS CloudOps Program - Batch 1: C...
Instances | EC2 | us-east-1
EC2 Instance Connect | us-east-1
Customer managed keys | KMS Cons
us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipv4&connType=standard&instanceId=i-069fc29741e2...
Search
[Alt+S]
United States (N. Virginia)
Account ID: 5187-3326-6010
user13
Transaction Summary
Install 2 Packages
Total download size: 5.8 M
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Is this ok [y/N]: y
Is this ok [y/N]: y
Downloading Packages:
(1/2): stunnel-5.58-1.amzn2023.0.2.x86_64.rpm 3.4 MB/s | 156 kB 00:00
(2/2): amazon-efs-utils-3.3.2-1.amzn2023.x86_64.rpm 52 MB/s | 5.7 MB 00:00
Total 36 MB/s | 5.8 MB 00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
Preparing : 1/1
Installing : stunnel-5.58-1.amzn2023.0.2.x86_64 1/2
Running scriptlet: stunnel-5.58-1.amzn2023.0.2.x86_64 1/2
Installing : amazon-efs-utils-3.3.2-1.amzn2023.x86_64 2/2
Running scriptlet: amazon-efs-utils-3.3.2-1.amzn2023.x86_64 2/2
Verifying : amazon-efs-utils-3.3.2-1.amzn2023.x86_64 1/2
Verifying : stunnel-5.58-1.amzn2023.0.2.x86_64 2/2
Installed:
amazon-efs-utils-3.3.2-1.amzn2023.x86_64 stunnel-5.58-1.amzn2023.0.2.x86_64
Complete!
[ec2-user@ip-172-31-8-173 ~]$ mkdir efs
[ec2-user@ip-172-31-8-173 ~]$
i-069fc29741e20a36c (FIRSTEC2-Rahul)
PublicIPs: 13.220.18.192 PrivateIPs: 172.31.8.173
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
Type here to search
28°C Partly cloudy
ENG
20:59
10-10-2026

1.8 Succesfully connected to firstec2 instance and mounted EFS filesystem with encryption mode as ON

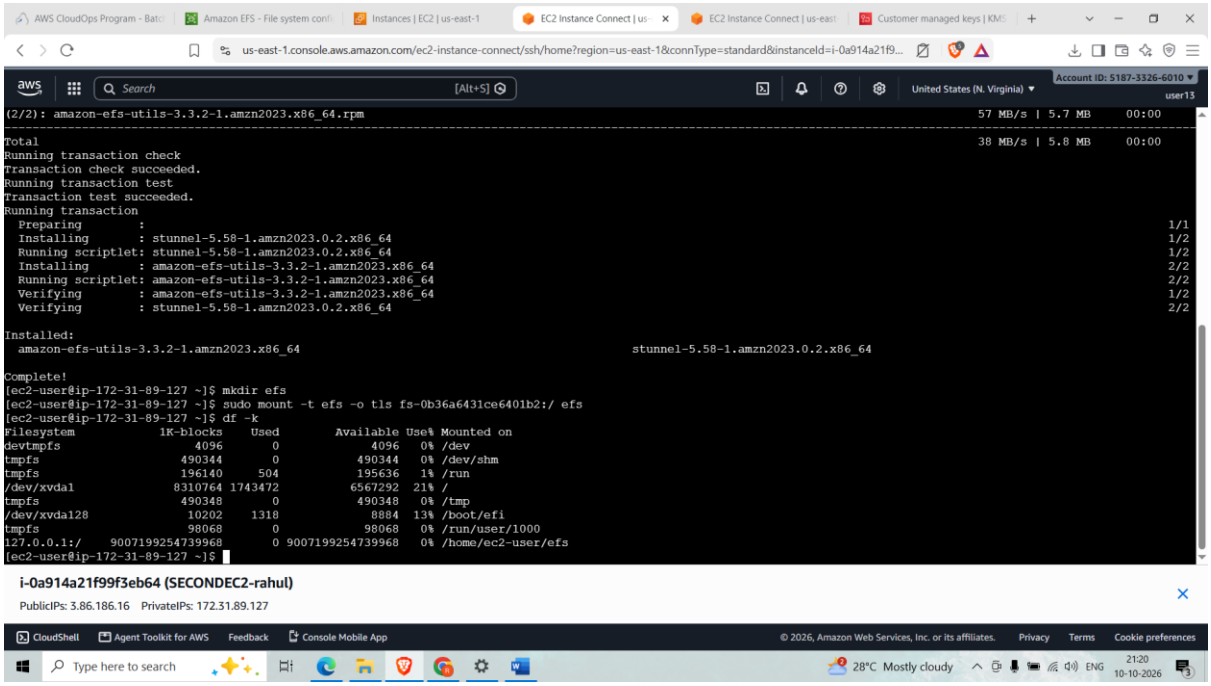


1.9 Steps to connect secondec2 instance and mounted EFS filesystem with encryption mode as ON

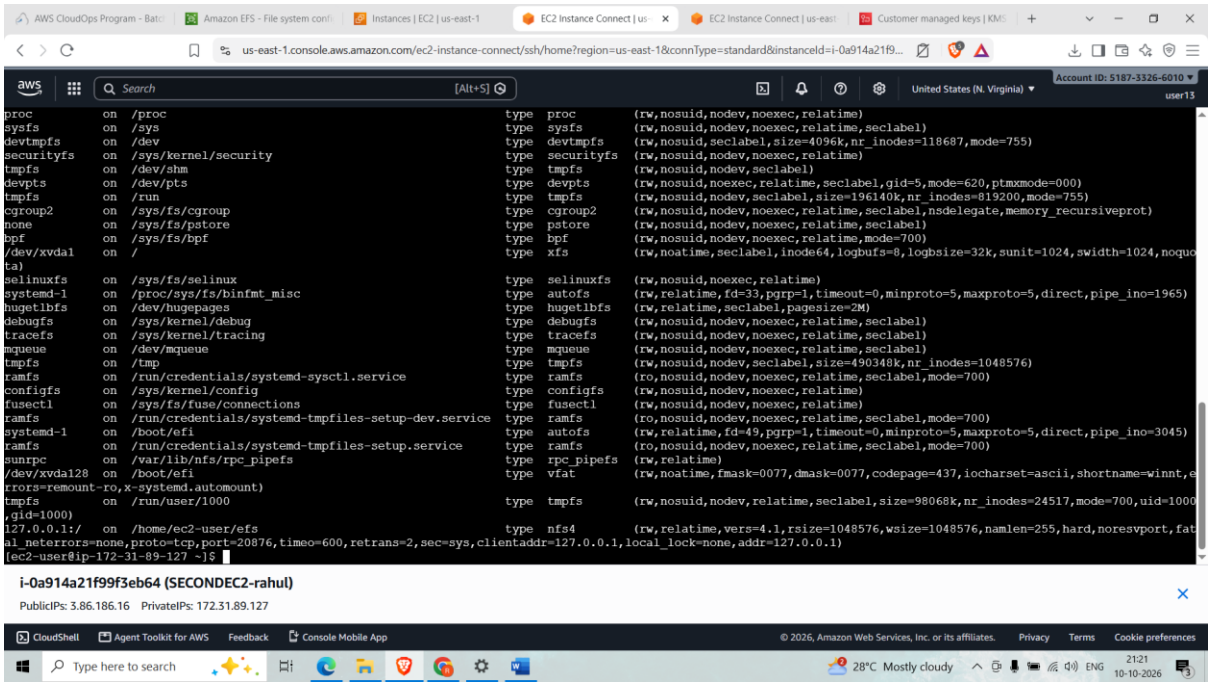
Connected the first ec2 instance (SECONDEC2-Rahul



Enter the following commands $ sudo yum install amazon-efs-utils $ mkdir efs

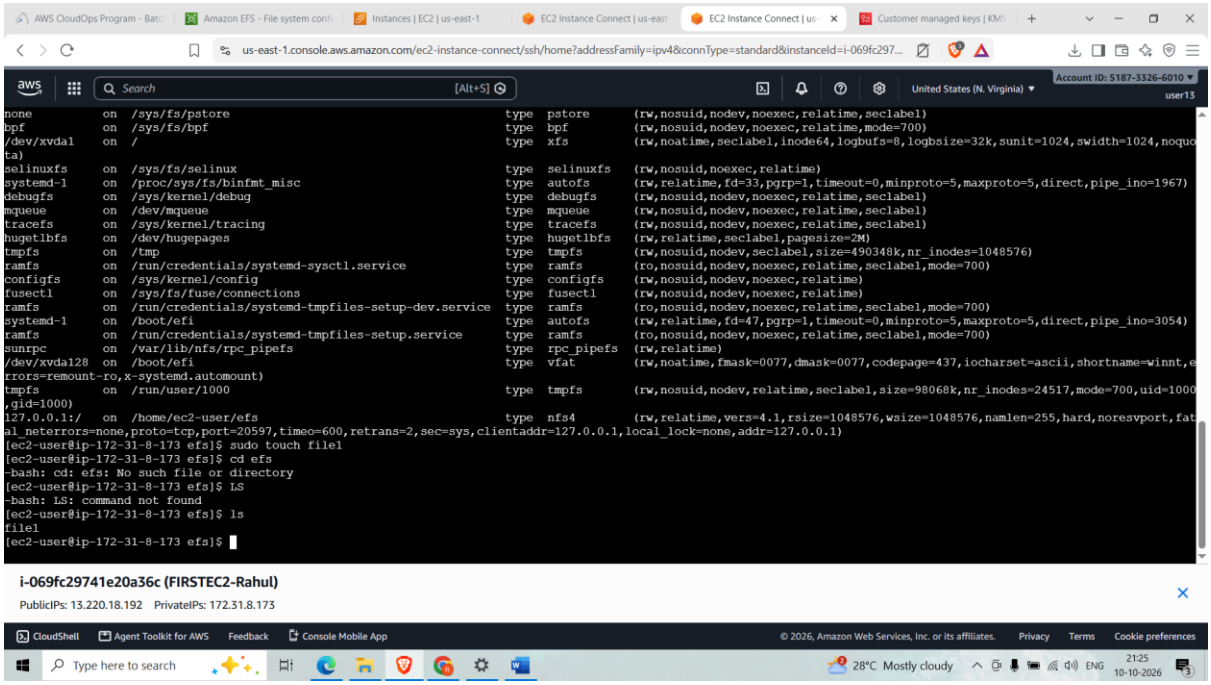


1.9 Succesfully connected to firstec2 instance and mounted EFS filesystem with encryption mode as ON

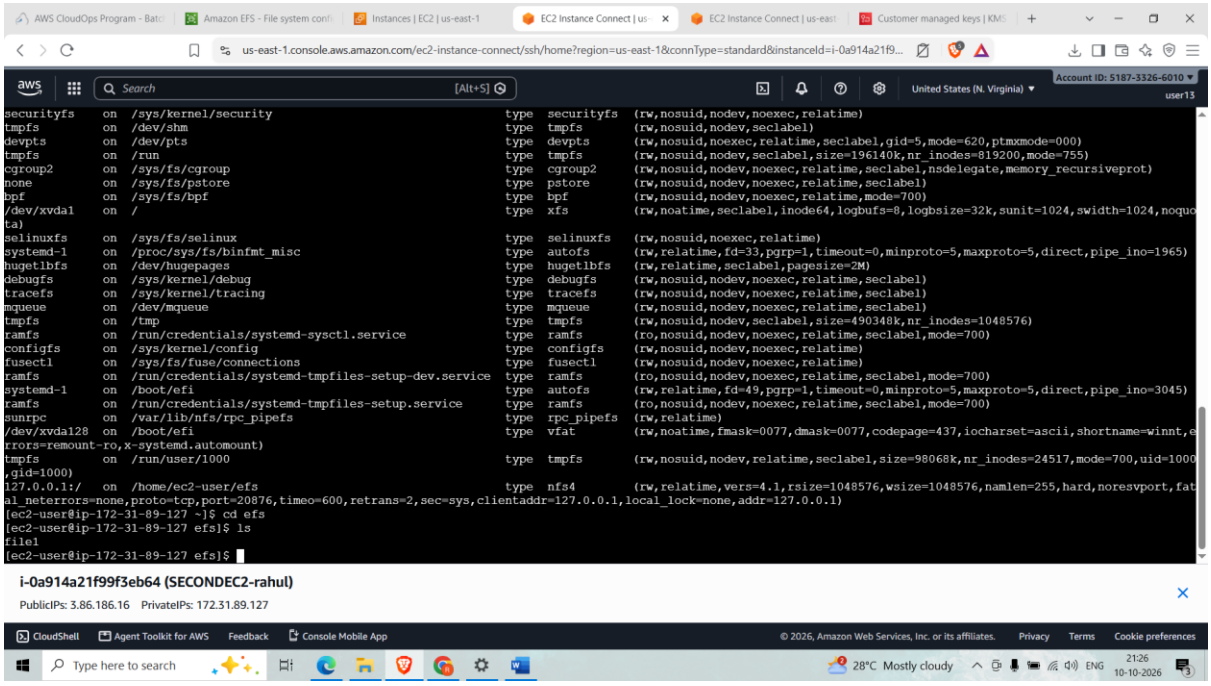


2.0 LETS CREATE A FILE IN FIRST INSTANCE and THIS FILE HAS TO BE VISIBLE IN SECOND INSTANCE

2.1FIRST INSTANCE , ADDED THE FILE NAMED FILE1



2.2 THE FILE IS VISIBLE TO SECOND INSTANCE AS WELL



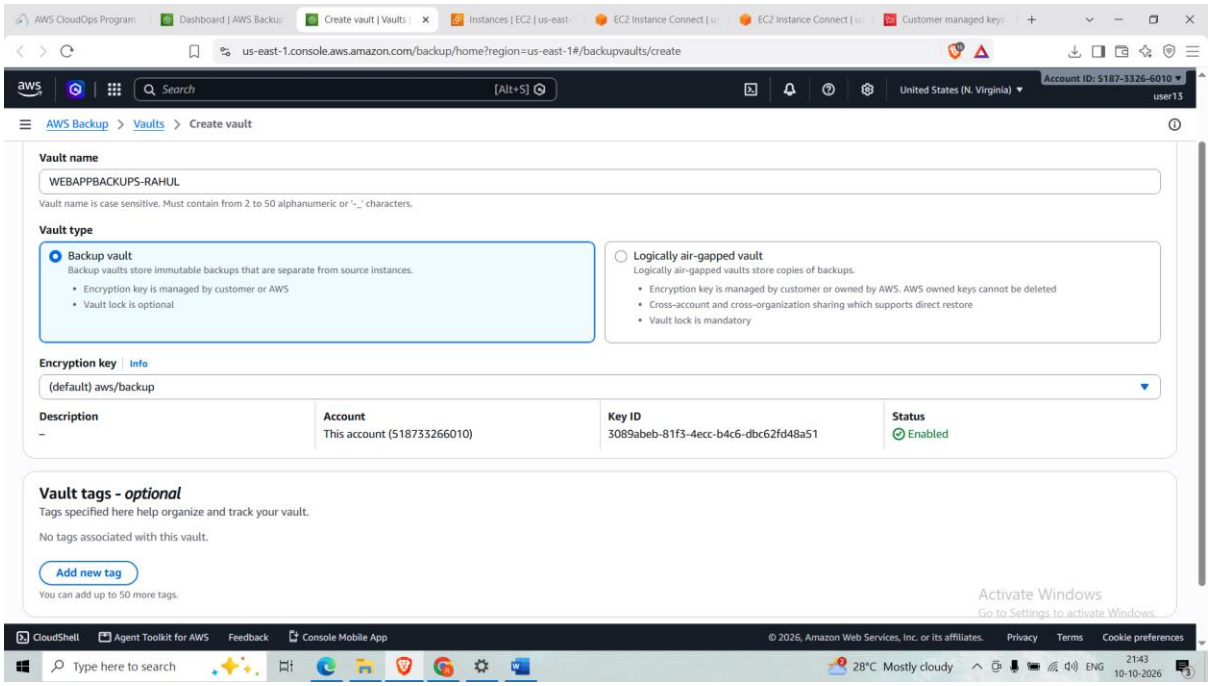
Now I can access file1 created by FIRSTEC2 through SECONDEC2. Hence, shared file system is working with both EC2

2nd CASE STUDY – WE WILL BACKUP THE FILES FOR THE RESOURCES USED IN CASE 1

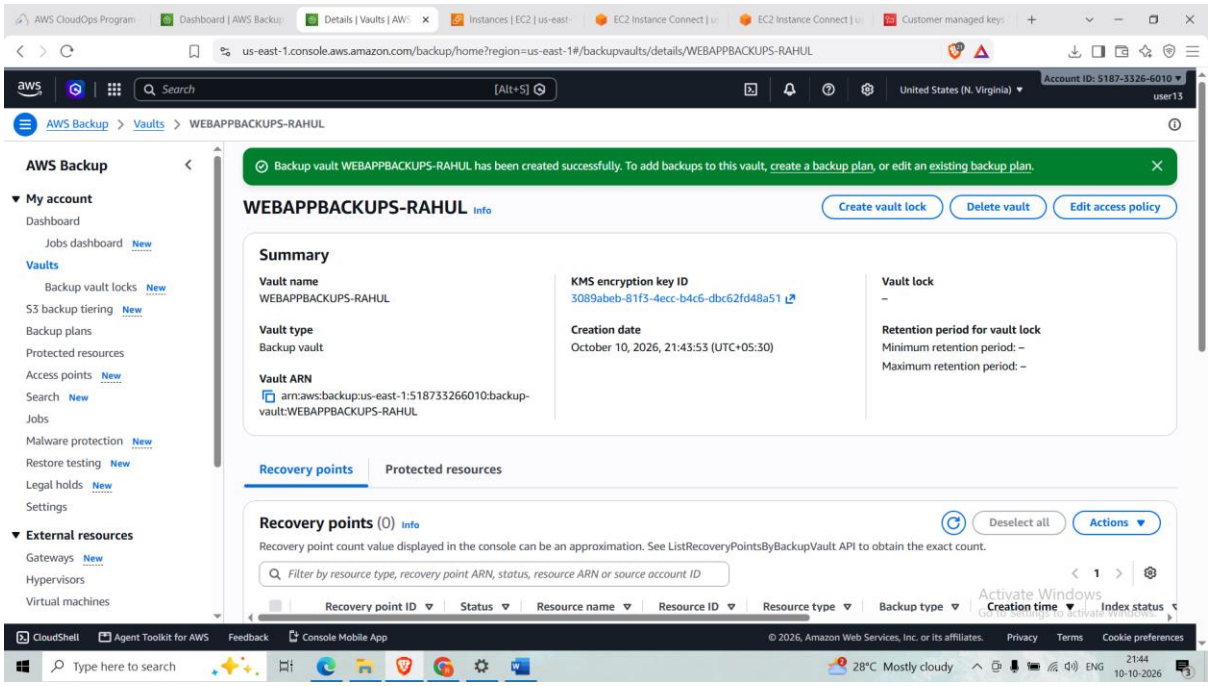
TOPIC : Backup Strategy to ensure mission critical data is being backed up regularly

TASK1: CREATING AWS Backup On-Demand

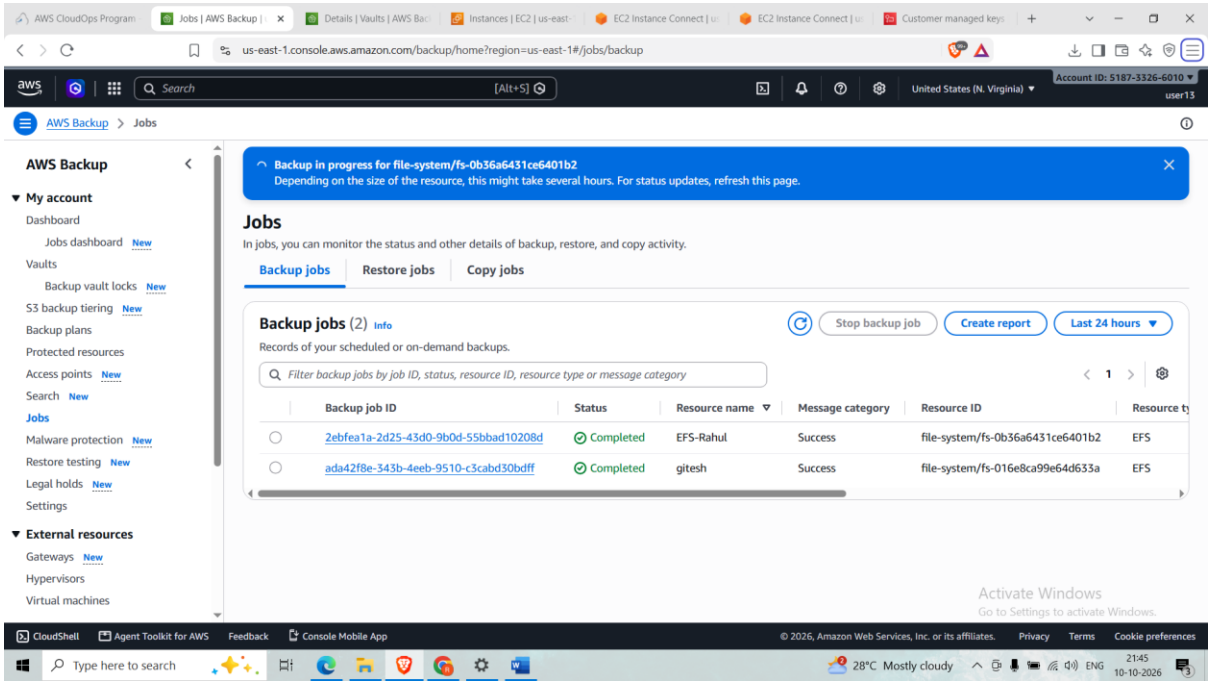
STEP 1.1 CREATING OOWN BACKUP VAULT WITH THE DEFAULT AWS KEY



OWN BACKUP VAULT CREATED

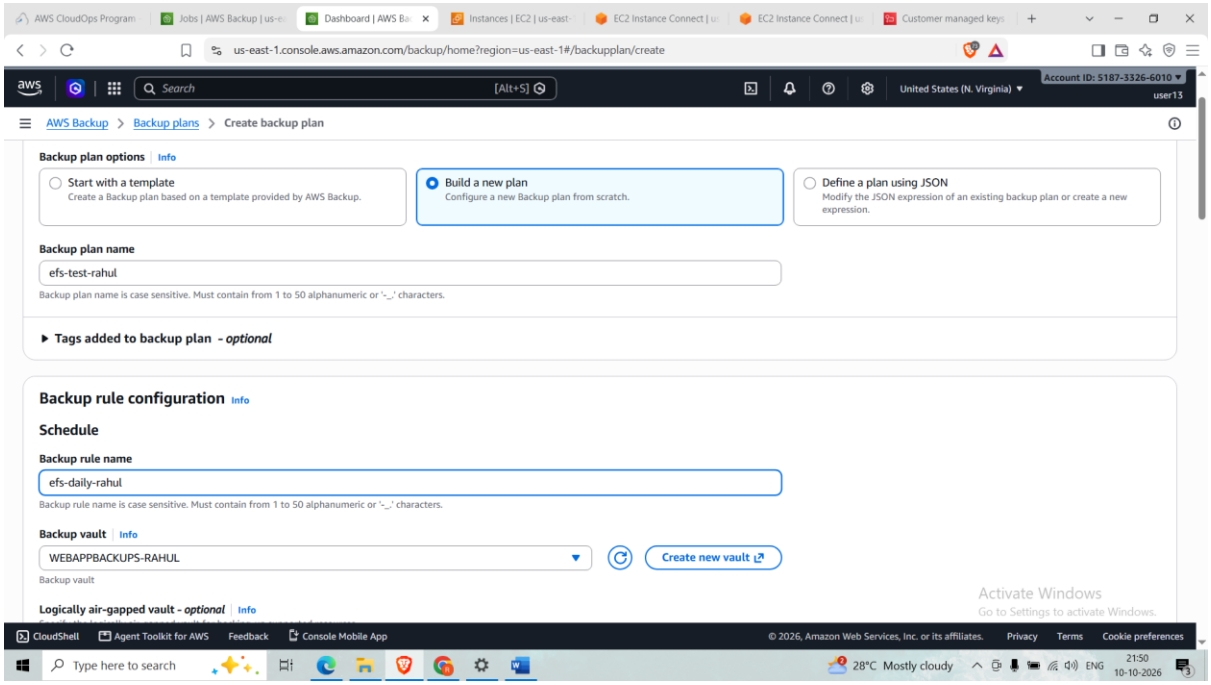


AWS On-Demand Backup has been successfully created

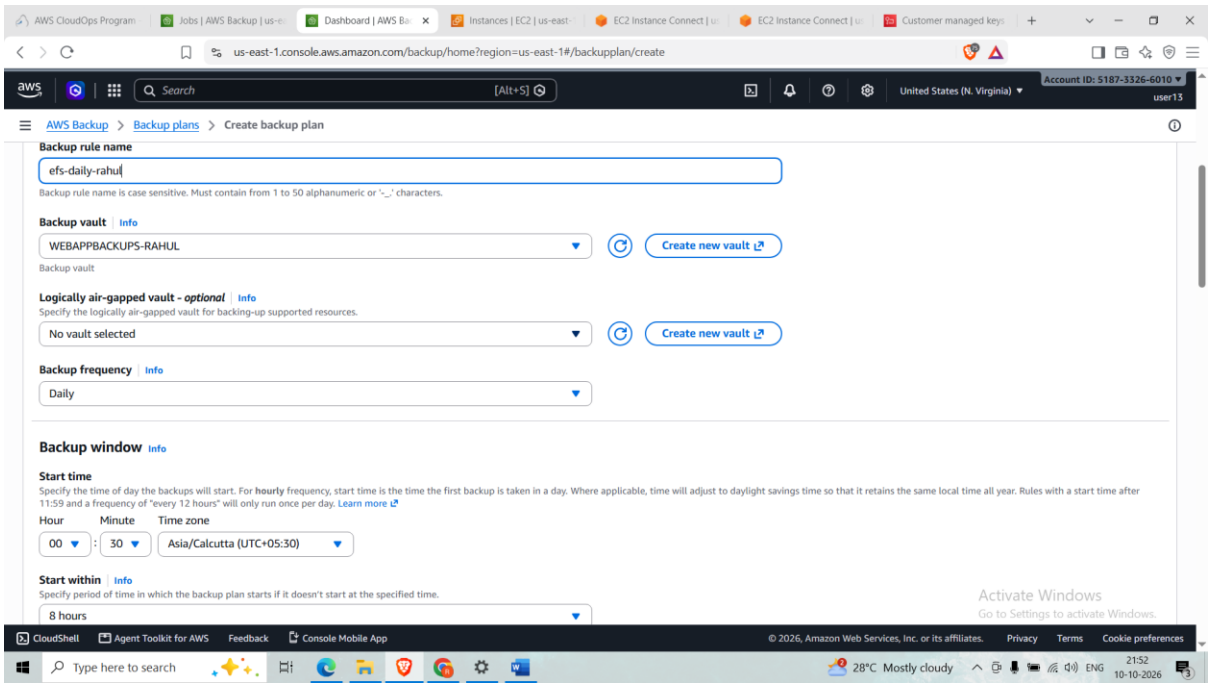


Task 2: For Configuring Automatic AWS Backup job of an Amazon EFS file systems.

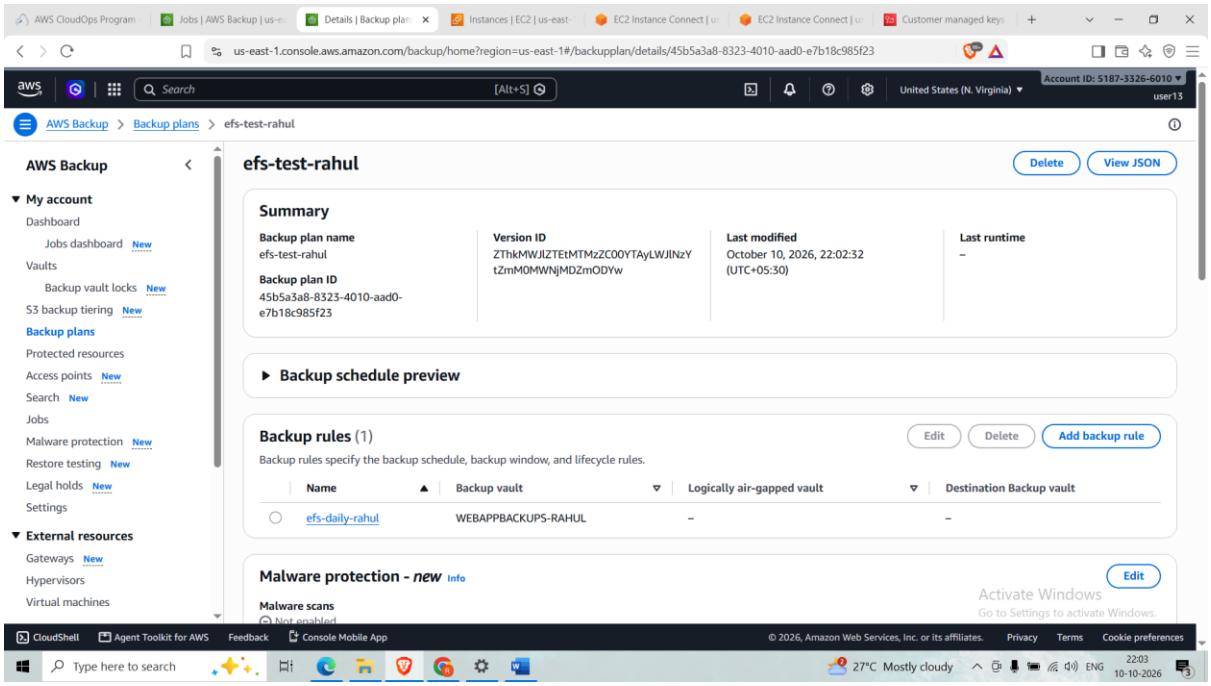
2.1 creating a backup plan with the backup vault we manually created



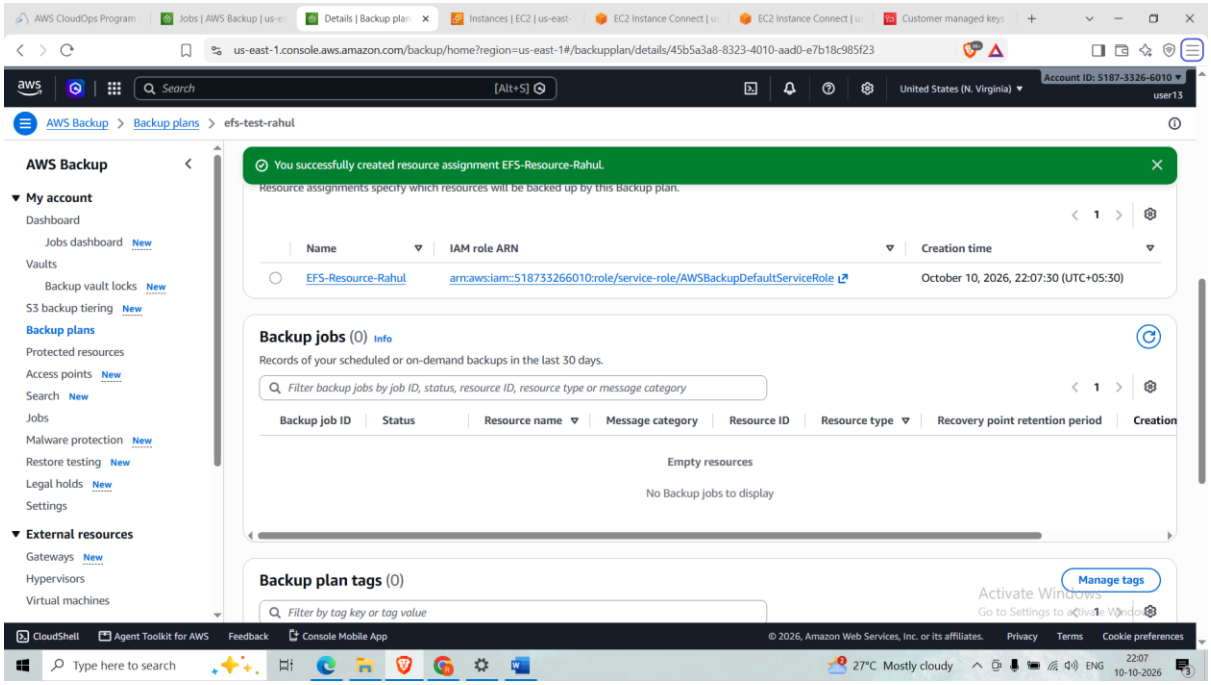
2.2 backup plan with daily frequency



2.3 successfully created backup plan and backup rules as well

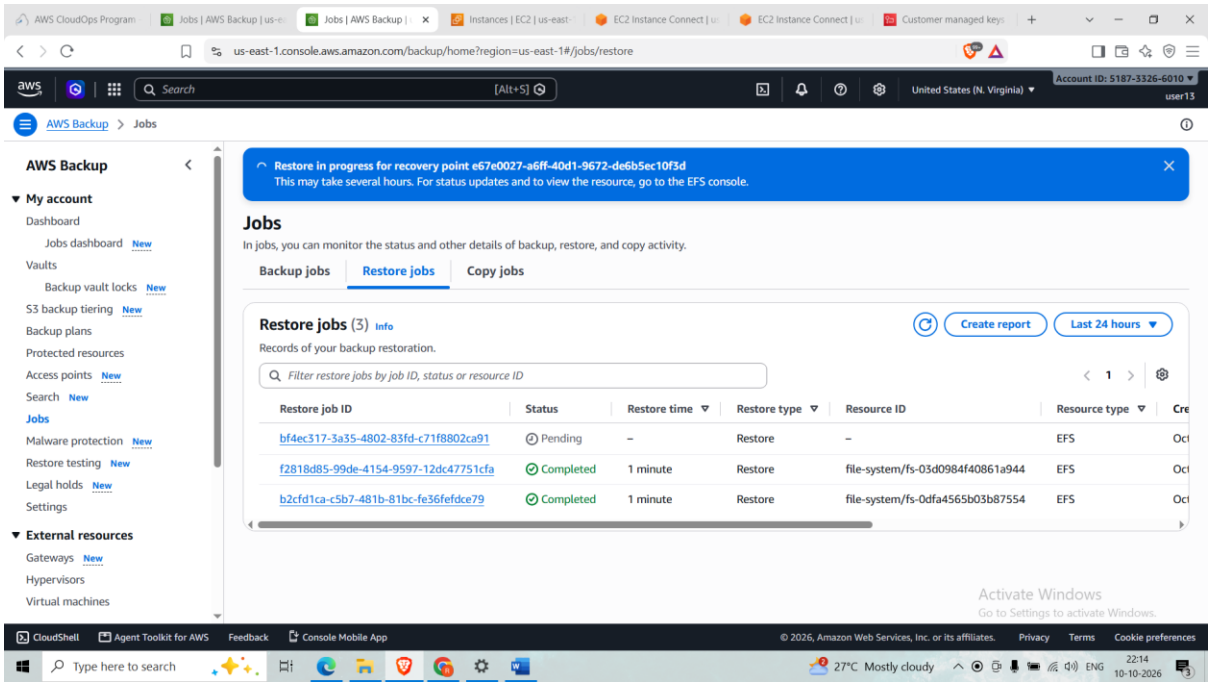


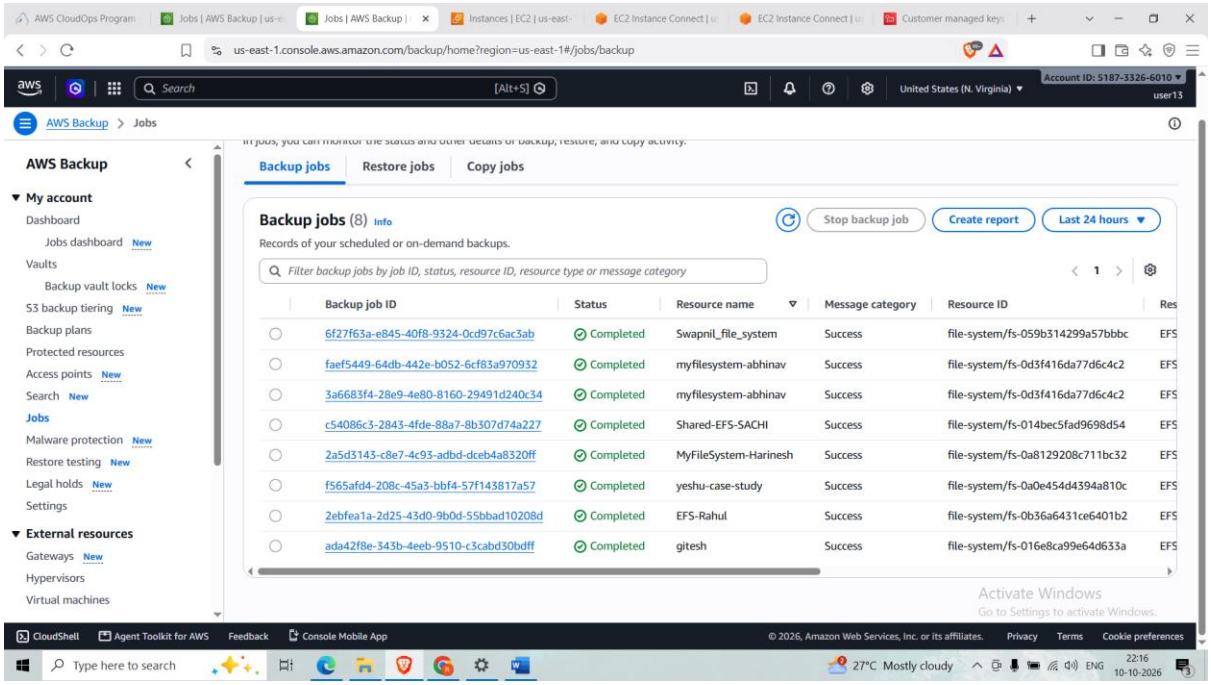
2.4 successfully added resource to automatic backup plan



AUTO BACKUP PLAN SUCCESSFULLY CREATED – WITH RESOURCES ADDED

Task 3: Restoring From Backups to ensure there is no data recovery issues.





SUCCESSFULLY RESTORED FROM BACKUP